

Wireless Water-Level Indicator

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Using this system, you can remotely monitor the water level of an overhead tank that is placed up to 30 metres away. The system features an RF transmitter-receiver pair, doing away with the need to run wires from the roof to ground. The transmitter is placed near the tank with sensors inside the tank to monitor the level of water. The sensed level is streamed wirelessly through the RF transmitter. This is received by the receiver unit placed remotely and decoded to indicate the water level on an LCD. It also has a buzzer that beeps when the water level drops below one-fourth level or when the tank is about to overflow.

When the tank is quarter-, half- or three-quarters-full, the percentage of the water level is flashed on the LCD. The system is developed using an EFY-KnS 8051 development board which is available from EFY associates Kits'n'Spares (KnS) for Rs 500.

EFY-KnS' 8051 development board

Fig. 1 shows the block diagram of EFY-KnS 8051 development board. It consists of a 40-pin IC socket, four 8-pin berg-strip male connectors, 9-pin serial COM port connector, MAX232 driver, 16-pin connector for the LCD and 5V regulator. Some other features of the board include:

1. Power LED with an on/off switch
2. Reset LED with a reset switch. This LED is also used as a program-status indicator.
3. A 5-pin male connector (not shown here) for 5V DC supply. The voltage is regulated to 5V using voltage regulator IC 7805.
4. Eleven jumper wires (16cm each) for connecting the LCD

Data and control pins of the LCD are not permanently connected to the microcontroller. So you can use any of the controller ports to connect the LCD using jumper wires.

Circuit description

Fig. 2 shows the circuit of the transmitter. It operates off 5V DC and consists of a sensor assembly, encoder HT12E



(IC1) and RF transmitter module (TX1). The sensor assembly consists of four BC547 npn transistors (T1 through T4), each connected to a water-level-sensor metal strip corresponding to one of the

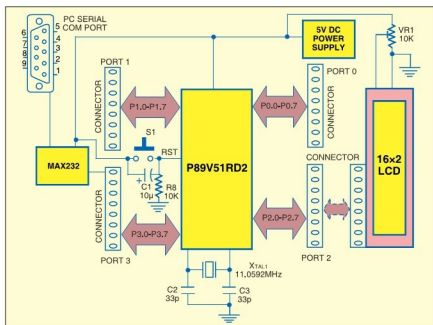


Fig. 1: Block diagram of EFY-KnS 8051 development board

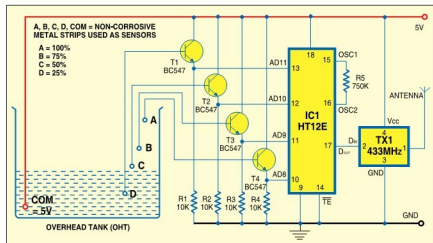


Fig. 2: Transmitter circuit